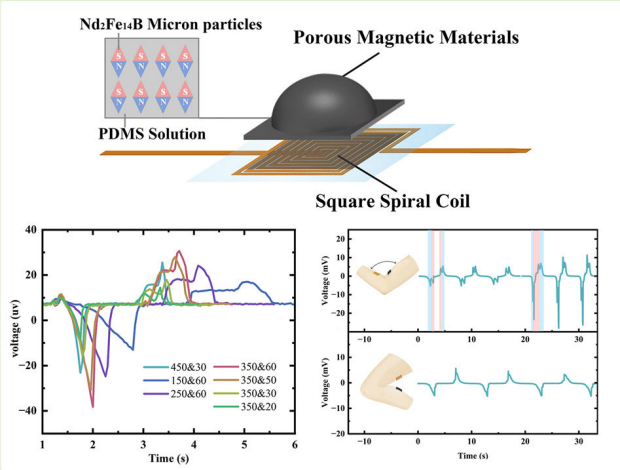
Self-Powered Pressure Sensor Based on Flexible Permanent Magnet Material

Chengxi Guo, Ziyang Fan, Yitao Ren, Yian Cheng, Yuan Jiang, Qinyan Zhang, Yifeng Li, Jianqiu Huang, and Huiyang Yu

***Abstract*—This work presents a self-powered pressure sensor based on flexible permanent magnet material. The sensor consists of a square spiral coil and magnetic material, which operates on electromagnetic induction. The flexible magnetic material is developed with the foaming agent ammonium bicarbonate (NH_4HCO_3). The magnetic characteristics and the mechanical properties of the magnetic component are greatly related to the ratio of NH_4HCO_3 and neodymium-iron-boron ($Nd_2Fe_{14}B$) doped in polydimethylsiloxane (PDMS). A semicircular hollow structure is proposed to improve the sensitivity and linearity of the sensor. This sensor is sensitive to dynamic forces and with high output voltage and fast response speed when the force is applied. The way that the force is applied affects the amplitude and the rising time of the voltage pulse generated in the sensor due to changes in the magnetic field created by flexible magnetic materials. Additionally, a portable testing system is built for performance testing. To meet the requirement for noncontact testing, the sensor can be separated into two components. A sensor array with a semicircular hollow shape has been constructed and can be worked as a counter. These applications indicate that this sensor can be used to identify human movement.**



***Index Terms*— Electromagnetic induction, flexible pressure sensor, magnet material, porous, self-powered, semicircular hollow structure.**

I. Introduction

With the advancements in manufacturing technology and network information technology, sensors have experienced rapid development toward miniaturization [1], networking [2], and intelligence [3]. However, as sensors become more commonly employed, they are not only occurring in factories and manufacturing but also being employed in medical [4], robotics [5], wearable electronics [6], and other industries. Selecting the appropriate sensor for various applications holds significant importance. Conventional sensors are resistive, capacitive [7], piezoelectric [8], etc. Capacitive and resistive pressure sensors have restricted applications in certain situations and need continuous power supply and maintenance. Piezoelectric sensors get over the power supply problem and have the ability to be self-powered, but many of them are difficult and costly to make. Recently, sensors made of magnetic materials and flexible substrates have been developed gradually since the development of micro- and nanomagnetic materials [9]. Compared to conventional sensors, magnetic sensors are more suited for usage in the automotive, medical, consumer electronics, and communications sectors [10]. Based on the piezomagnetic principle [11], Hall effect [12], electromagnetic induction principle [13], [14], and the magnetostrictive effect [15], there is a wide demand for developing magnetic sensors.

Magnetic materials have the advantages of chemical stability, biocompatibility, superior magnetic moment [16], flexible electronic devices variable shape, lightweight, and reconfig-

Received 18 August 2024; revised 18 September 2024; accepted 18 September 2024. Date of publication 27 September 2024; date of current version 14 November 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62104105 and Grant 62271144 and in part by the National Students' Platform for Innovation and Entrepreneurship Training Program under Grant 202310291041Z and Grant 202310291215Y. The associate editor coordinating the review of this article and approving it for publication was Prof. Kai Wu. *(Chengxi Guo and Ziyang Fan contributed equally to this work.) (Corresponding author: Huiyang Yu.)*

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Digital Object Identifier 10.1109/JSEN.2024.3465664

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urable function subvert the stiff physical form of conventional electronic systems [17]. For example, magnetically driven microrobots exhibit advantages when contrasted with other driving strategies (such as electricity, heat, and light) [18], [19], [20] and can also perform remote wireless operations, which are practical in the field of minimally invasive surgery [21]. As for sensors, signal can be transmitted without physically touching the target object as the magnetic field can pass through a wide variety of nonmetallic materials [22]. As flexible magnetic material can tolerate bending, stretching, and other mechanical deformations, it has the possibility of miniaturization and integration, both in the form of sensors or actuators. The properties of magnetic sensors include high measurement sensitivity, consumption, robustness, and low power [23].

For flexible magnetic sensors, the pressure and magnetic field variation is related. The pressure applying on the sensor leads to variations in the magnetic field surrounding it. For example, Miller et al. [11] utilized the magnetoelastic effect of magnetic polymers and developed a magnetic sensor with a magnetic cylindrical structure nested in pure polydimethylsiloxane (PDMS) substrate. Man et al. [24] fabricated a haptic sensor with a magnetic cilia structure, in which the top is flexible magnetic material and the bottom is pure PDMS. Utilizing the varying magnetic field and electromagnetic induction, a self-powered sensor can be developed by combining flexible material with a spiral coil. Mo et al. [14] proposed a tactile sensor, in which an elastomer embedded in a permanent magnet is combined with an array of inductive coils for slip detection. Zhou et al. [13] developed a biomimetic cilia electronic skin. The mixture of PDMS, carbonyl iron particles, and neodymium-iron-boron ($Nd_2Fe_{14}B$) was sprayed and magnetized on the exterior of the cilia structure.

The structure of the flexible magnetic component is crucial for optimizing the performance of the sensor. Recently, numerous investigations have been carried out on the structure of magnetic components for various corresponding scenarios, including cilia [25], pyramid [26], porous [27], and others. For instance, Zhou et al. [27] prepared a soft system consisting of micromagnets and a porous silicone rubber matrix. This porous structure not only reduces Young's modulus but also provides a small space for micro- or nanomagnetic materials to reorient and move during the process of pulsed magnetization. In addition, it is important to choose a suitable flexible substrate. Flexible substrate when combined with $Nd_2Fe_{14}B$ will affect its properties, such as magnetism and modulus [28], [29].

According to the published works reported recently, it is evident that improving pressure sensor performance by utilizing flexible magnetic materials is effective. The structure with embedded magnet is unsuitable for scenarios needing high elasticity due to the nonuniform magnetic field and mechanical properties of the material. The cilia or thin film structures are limited in measurement ranges. Moreover, sensors based on electromagnetic induction are highly sensitive to dynamic pressures, and the output voltage is more conducive to detect the varying speed of pressure change.

In this article, a flexible permanent magnet pressure sensor is developed, which is based on the electromagnetic induction principle. Based on the previous research experience of our group on porous materials [30], the pressure-sensitive property of the magnetic material was improved by designing the component and the structure of the magnetic material. $Nd_2Fe_{14}B$ and PDMS were utilized as the magnetic materials and substrates, respectively. The effects of porosity, magnetic powder content, and structure on the performance of the sensor were investigated.

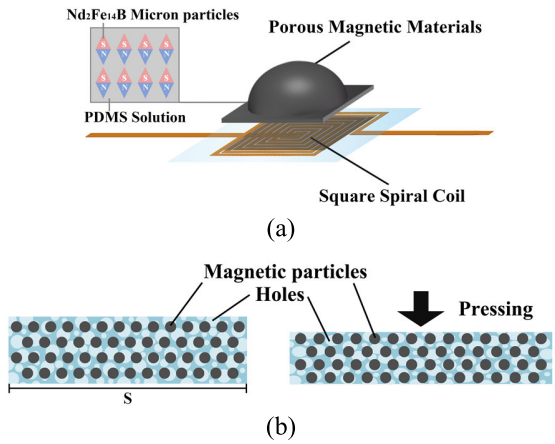

Fig. 1. (a) Structure of the flexible magnetic pressure sensor. (b) Status change of the flexible magnetic material before and after compression.

II. Methods

A. Structure of the Sensor and Its Working Principle

The pressure sensor consists of two parts: a pressure-sensitive magnetic component and a square spiral coil. The structure of the sensor is shown in Fig. 1(a). The center of the coil is located below the magnetic component. When pressure is applied on the magnetic component, the magnetic flux through the coil enlarges and a voltage pulse is generated in the coil. Once the pressure is released, the magnetic flux passing through the coil decreases and generates a reverse voltage pulse within the coil. As a result, the strength and speed of the pressure can be measured according to the amplitude and cycle of the induced voltage generated in the coil. Fig. 1(b) illustrates the working principle of the sensor. Since the area of the material expands and the density of magnetic particles increases when the porous magnetic material is pressed, the material exhibits a stronger magnetic field when pressed. The theoretical basis of the working mechanism of the sensors is based on Faraday's law of electromagnetic induction principle. The equation of the induced voltage is as follows:

$$E = -N\frac{d\boldsymbol{\Phi}}{dt} = -NS\frac{d\boldsymbol{B}_z}{dt}. \tag{1}$$

Here, E is the induced electromotive force, N is the turns of the coil, $d\boldsymbol{\Phi}/dt$ is the magnetic flux variation ratio to time, $\boldsymbol{\Phi} = \boldsymbol{B} \times S$, $\boldsymbol{B}$ is the magnetic flux density in the direction perpendicular to the coil, and S is the area of the coil covered by the magnetic field. Electromagnetic induction is a phenomenon where the generation of an electromotive force or voltage occurs in a conductor when there is a change in the magnetic field around it [31]. According to (1), the induced electromotive force is proportional to the ratio of magnetic flux

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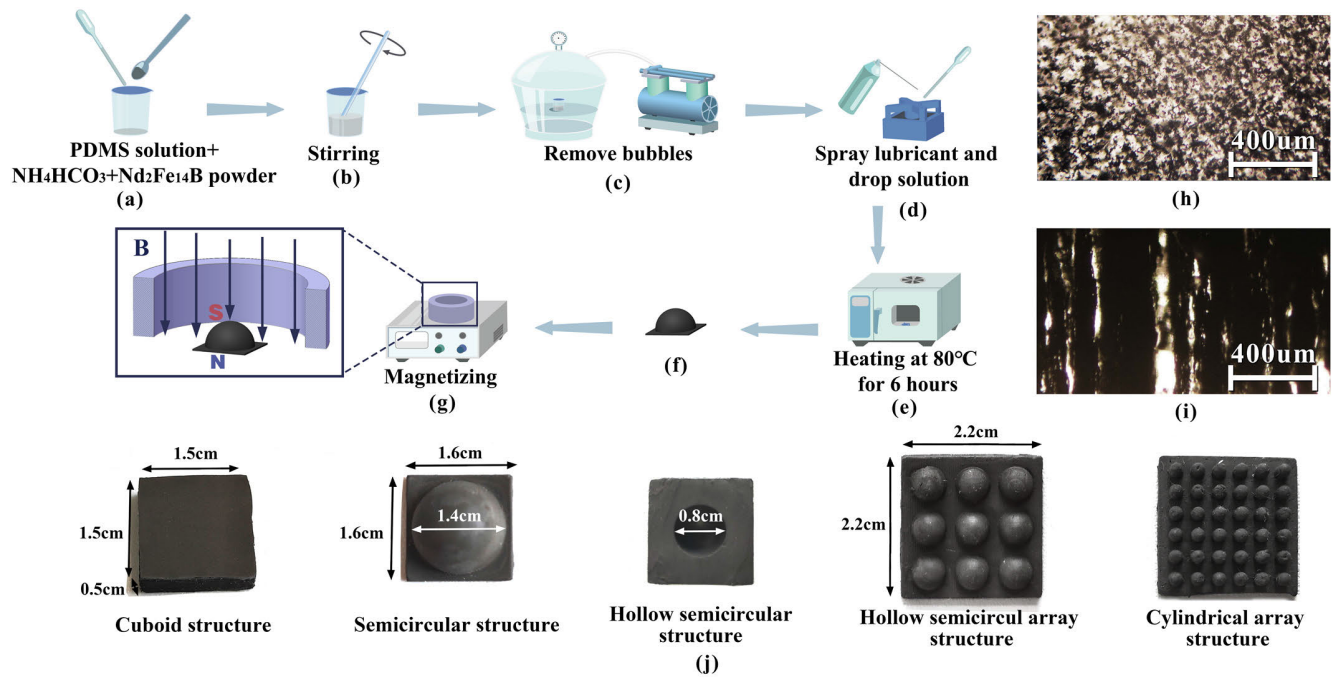

Fig. 2. (a)–(g) Fabrication process of magnetic materials. (h) Distribution of magnetic particles in materials using the cure-magnetizing method under a microscope is observed. (i) Distribution of magnetic particles in materials using the magnetize-curing method under a microscope is observed. (j) Different structures of the magnetic components.

TABLE I
PARAMETERS OF $Nd_2Fe_{14}B$ POWDER

Parameters	$Nd_2Fe_{14}B$
Average radius	5μm
Density	7.61 g/cm^3
Residual magnetic induction intensity $\mathbf{B}_r$	875-905 mT
Maximum magnetic energy product $(\mathbf{BH})_{max}$	102-118 kJ/m^3
Intrinsic coercivity $\mathbf{H}_{ci}$	530-610 kA/m
Coercivity $\mathbf{H}_c$	440 kA/m
Maximum operating temperature	100℃

variation through the square spiral coil. There are many factors related to E. The sensor described in this article is capable of generating voltage upon pressure application, attributed to alterations in $d\boldsymbol{B}_z/dt$, which means that E is determined by the variation range of the magnetic field and the duration of these variations.

B. Fabrication Process of Coil and Magnetic Component

The fabrication of the sensor in this article consists of two steps: coil printing and the magnetic component fabrication. The coil is printed on a sticker machine using copper foil. $Nd_2Fe_{14}B$ (D50, Tianjin Magnequench Company Ltd.) is used to fabricate the magnetic component of the sensor, and some of its parameters are shown in Table I.

$Nd_2Fe_{14}B$ powder is magnetized using a long pulse magnetizer (J311, Nanjing Golden Wolf Electric Company Ltd., China), which is capable of generating a magnetic field close to 1 T. The whole magnetization process can be finished in a few seconds. PDMS (SYLGARD[1] 184 Silicone Elastomer Kit, Corning Inc.) is employed as the flexible substrate. Ammonium bicarbonate (NH_4HCO_3) is used to fabricate porous materials. The detailed fabrication process of the magnetic component is shown in Fig. 2(a). The $Nd_2Fe_{14}B$ powder and NH_4HCO_3 are added to the liquid PDMS solution. The mixture is thoroughly mixed by hand and stir for 5 min [Fig. 2(b)]. The mixture is put into the vacuum chamber, and an electric pump is utilized to extract air in the chamber [Fig. 2(c)]. This step is meant to get rid of air from the PDMS and magnetic particle mixture. A mold is prepared with a 3-D printing machine Fig. 2(d). Some lubricant is sprayed on the mold surface to reduce the difficulty of demolding, and then, the PDMS mixture is dropped into the mold. The mold is put into the oven and heated at a temperature of 80 °C for 6 h [Fig. 2(e)]. The magnetic material is demolded from the mold [Fig. 2(f)]. To magnetize the magnetic material, it is put into a pulse magnetizer [Fig. 2(g)]. The temperature 80 °C is utilized for two reasons. First, the heating temperature cannot be very low. NH_4HCO_3 can undergo smooth decomposition at a temperature of 80 °C. Second, the temperature cannot be very high which will affect the magnetic material. The 80 °C temperature has no obvious effect on the magnetic performance of $Nd_2Fe_{14}B$.

The distribution of magnetic particles in the microscope is depicted in Fig. 2(h). The particles within the substance are uniformly distributed. Fig. 2(i) illustrates the distribution of magnetic particles using a magnetization-curing way, which is another method of fabricating magnetic material. The distribution of magnetic particles forms chains. In this method, the mixture is magnetized before the PDMS is cured, then, the uncured magnetic mixture is put into the specified mold and two permanent magnets are placed at the side of the mold

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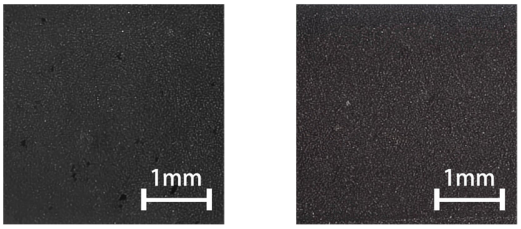

(a) (b)

Fig. 3. Cross section of magnetic components with different NH_4HCO_3 concentrations. (a) 0 wt%. (b) 0.5 wt%.

to control the distribution of magnetic particles, and finally, the magnetic material is obtained after being cured in the oven. In contrast to the method displayed in Fig. 2(a)–(g), this magnetization-curing method has two shortcomings. First, the distribution and orientation of the chain made using this method are completely determined by the magnetic field of the permanent magnet used in curing. Second, this chain structure makes the material anisotropic to the force [32]. When a force is exerted no matter in an axial or in radial direction, the material will exhibit strain in corresponding directions. It is more difficult to generate strain under axial force. Hence, the magnetization-curing method reduces the sensitivity of the sensor. These problems did not occur in magnetic material prepared by the curing-magnetization method. Hence, the curing-magnetization method used in our experiment when fabricating flexible magnetic materials is better.

Fig. 3 shows the cross-sectional views of the magnetic components with and without NH_4HCO_3 using the curing-magnetization method. It can be seen that the magnetic material doped with 0.5 wt% NH_4HCO_3 has more small holes in the interior than the one without NH_4HCO_3 doped, proving that NH_4HCO_3 is effective as a vesicant.

Since the strain of the magnetic component is greatly related to the change in the magnetic flux in the coil, the magnetic components are designed with structures that can create an obvious change in the magnetic field when pressed. In order to study the relationship between the strain of the magnetic component and the composition or structure of the magnetic materials, cuboid structure (1.5 × 1.5 × 1.5 cm), semicircular solid structure (square side length = 1.6 cm and outer radius = 0.7 cm), semicircular hollow structure (internal radius = 0.4 cm), semicircular hollow structure array (square side length = 2.2 cm), and cylindrical structure array are designed and illustrated in Fig. 2(j).

III. Results and Discussion

A. Mechanical Properties and Magnetic Field Studies of the Sensor

The testing system of the sensor is shown in Fig. 4. A force-detecting machine is used to apply and measure force, as well as test the deformation. A digit multimeter is connected to two ends of the copper coil in order to measure the voltage generated in the coil when pressure is applied.

The mechanical properties of different magnetic components are shown in Fig. 5. Fig. 5(a) shows the strain of cuboid magnetic components doped with different weight

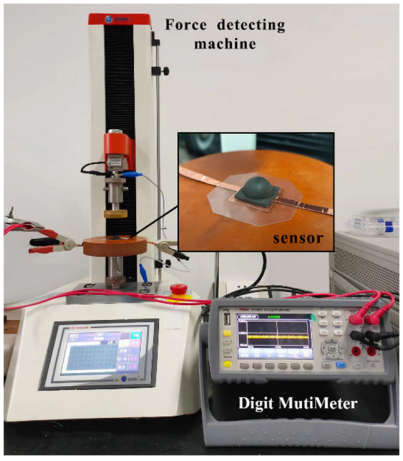

Fig. 4. Connection testing system consists of two parts: a force-detecting machine and a digit multimeter.

ratios of $Nd_2Fe_{14}B$. Strain is calculated by dividing the deformation L by the sensor height. The testing results indicate that there is a negative correlation between the weight ratio of $Nd_2Fe_{14}B$ and the strain, meaning that as the weight ratio of $Nd_2Fe_{14}B$ increases, the strain decreases. This is because the addition of magnetic particles enhances the hardness of the composite magnetic material. When the magnetic component is doped with 80 wt% $Nd_2Fe_{14}B$, the strain of the cuboid structure is about 12%. Moreover, it is difficult to demold the composite magnetic material from the 3-D-printed mold, and it is prone to fracture when subjected to a large force. Since 80 wt% $Nd_2Fe_{14}B$ of the material is fragile, the magnetic material doped 70 wt% $Nd_2Fe_{14}B$ is fabricated. Fig. 5(b) shows the strain of cuboid magnetic components doped with different weight ratios of NH_4HCO_3 but all doped with 70 wt% $Nd_2Fe_{14}B$. The strain of the material increases from 18% to 30.25% under the force range from 0 to 50 N. The testing results indicate that there is a correlation between the weight ratio of NH_4HCO_3 and the strain of the magnetic component. The higher weight ratio of NH_4HCO_3 results in a larger strain. This is because the doping of NH_4HCO_3 improves the porosity of the material. Fig. 5(c) shows the strain of magnetic components with different structures. Due to the area change of the semicircular construction during compression, the force (N) is judged instead of pressure (Pa). According to the testing results, the semicircular hollow structure exhibits a larger strain compared to the cuboid structure. Specifically, the strain of the semicircular hollow structure is 47.25% at 50-N force. All the structures in this measurement are fabricated the same material. This indicates that the hollow semicircular construction has a higher pressure-sensitive property.

A digital Tesla meter (TD8620, Changsha Tianheng Measurement and Control Technology Company Ltd.) is used to measure static and dynamic magnetic fields on the surface of magnetic components. The static magnetic field is carried out to test the magnetic field at the center of the upper surface of the component, specifically the top of the semicircular hollow structure. The measurement is carried out with the probe remain perpendicular to the bottom surface of the component.

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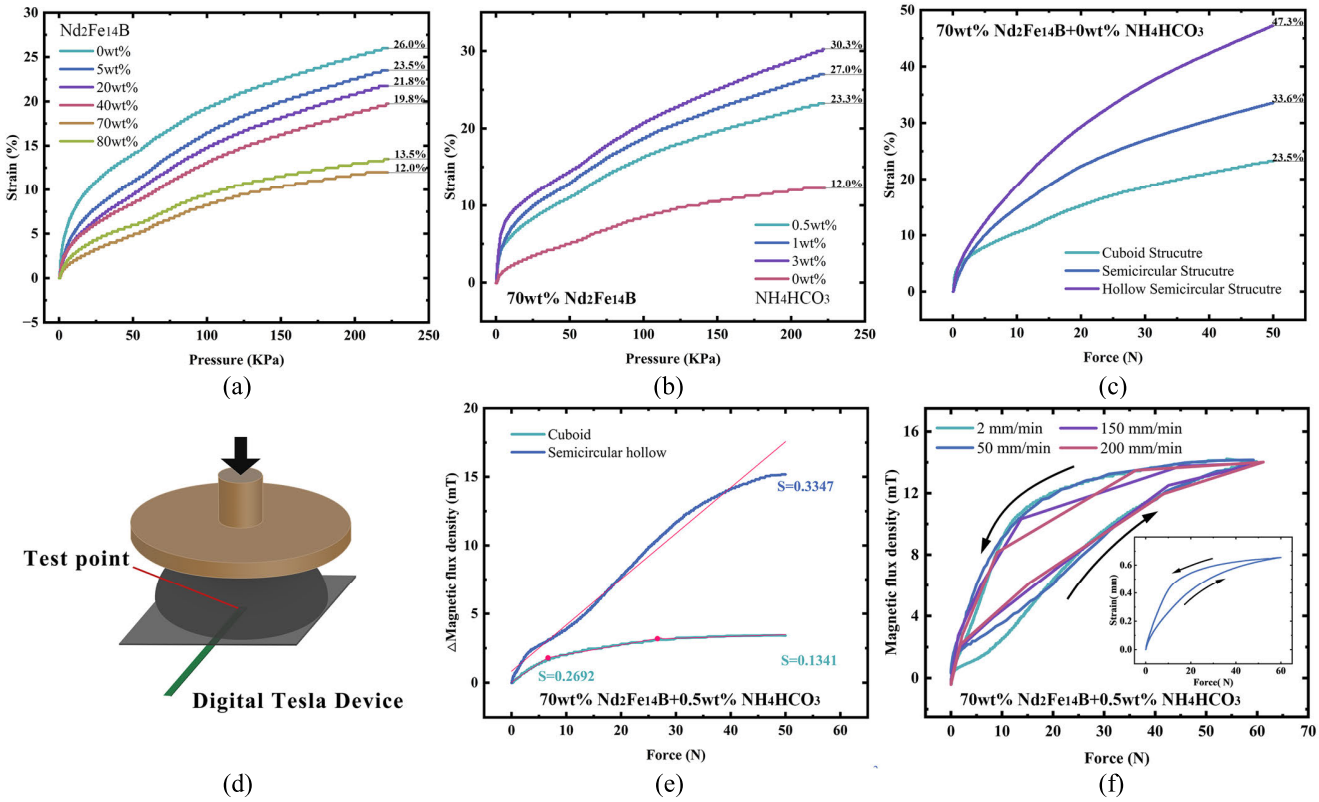

Fig. 5. Relationship between the strain of the magnetic component and the applied force is affected by (a) $Nd_2Fe_{14}B$ concentration, (b) NH_4HCO_3 concentration, and (c) structure of the magnetic component. (d) Test install for magnetic fields employs a digital Tesla meter to measure the magnetic field right below the sensor. (e) Magnetic field variation of cuboid and semicircular hollow structures under force range from 0 to 50 N. The sensitivity curve of the cubic structure is linearly fit into three segments, with the sensitivity decreasing from 0.2692 to 0.1341 mT/N. The curve of the semicircular hollow structure is fit to a line with a sensitivity of 0.3347 mT/N. (f) Force–magnetic hysteresis curves at various pressing speeds and the force–strain hysteresis curves were measured at a speed of 2 mm/min.

TABLE II
MAGNETIC FIELD ON THE UPPER SURFACE OF THE CUBOID STRUCTURE WITH DIFFERENT $Nd_2Fe_{14}B$ CONTENTS

$Nd_2Fe_{14}B$ (wt%)	2	5	10	20	40	60	70	80
Magnetic field (mT)	0.63	1.46	3.02	5.60	15.87	27.20	44.50	54.11

TABLE III
MAGNETIC FIELD ON THE UPPER SURFACE OF THE MAGNETIC COMPONENTS IN DIFFERENT STRUCTURES AND NH_4HCO_3 CONTENTS

NH_4HCO_3 (wt%)	$Nd_2Fe_{14}B$ (wt%)	Magnetic field (mT)	
		Cuboid	Semicircular hollow
0	70	44.5	90.02
0.5		42.13	87.12
2		38.49	64.98
3		32.02	59.46

Before magnetizing, the magnetic field on the surface of all magnetic components is smaller than 0.1 mT, indicating that they are not magnetic. After magnetizing, the magnetic field of the surface of the cuboid with 2 wt%∼80 wt% $Nd_2Fe_{14}B$ is shown in Table II. The magnetic field on the magnetic components surface can be intensified by doping higher concentrations of $Nd_2Fe_{14}B$ powder.

Table III shows the magnetic field detection results on the upper surface of the cubic and semicircular hollow structures with various doping concentrations of NH_4HCO_3. The doping concentration of $Nd_2Fe_{14}B$ in this experiment is 70 wt%.

As shown in Table III, with the increasing concentration of NH_4HCO_3, the magnetic field of magnetic components surface tends to decrease. This is because adding too much NH_4HCO_3 causes the holes to squeeze the space occupied by magnetic particles in the magnetic material. Consequently, the addition of NH_4HCO_3 to the substance can enhance its responsiveness to pressure; however, too much amount will diminish the magnetic characteristics of the magnetic component. In addition, the magnetic field strength on the surface of the semicircular hollow component is approximately twice as high as that of the cuboid structure when both are doped with the same volume of NH_4HCO_3, which is attributed to the curved surface of the semicircular structure. This implies that the cavity in the semicircular structure is more favorable to generate larger variation of the magnetic field.

The testing system of the dynamic magnetic field is depicted in Fig. 5(d). A digital Tesla meter is used to measure the change in the magnetic field directly below the material, and a force-detecting machine is used to apply force from 0 to 60 N. Since the magnetic field is a vector quantity, the digital Tesla meter mainly focuses on the magnetic field perpendicular to the bottom of the magnetic component. The magnitude of the magnetic field change in these two structures is tested and applied the same force. Fig. 5(e) displays the magnetic

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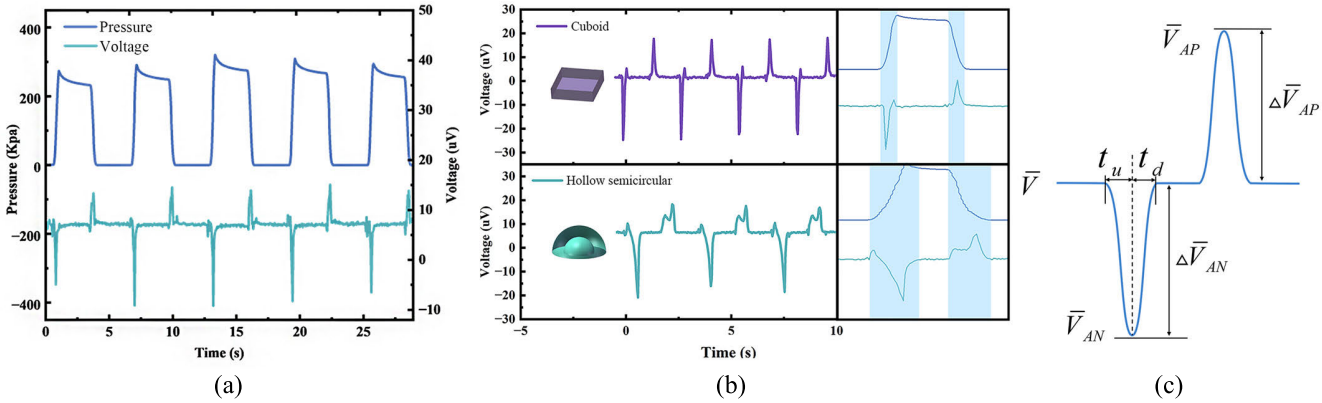

Fig. 6. (a) Voltage generated by the sensor with a cuboid structure component when pressure is exerted. (b) Cuboid and semicircular hollow structures generate voltage comparison, and a comparison is conducted between the force and voltage waveforms of the two structures. (c) Legend of an output voltage cycle.

field when the cuboid and semicircular hollow structures are subject to force. The variation of the magnetic field is just 3.5 mT under the force range of 0∼50 N, and its sensitivity gradually decreases as force increases, which is from 0.2692 to 0.1341 mT/N.

In contrast, the variation range of the magnetic field of the semicircular structure is four times greater than that of the cuboid structure. The semicircular structure is far more sensitive than the cuboid, with a sensitivity of 0.3347 mT/N. Semicircular hollow structures can generate a larger variation of the magnetic field due to the gradual movement of the top of the structure, which has a strong magnetic property when pressed toward the bottom. This movement results in a greater change in the magnetic field. As a result, the semicircular hollow structure is better than the cuboid when used in a pressure sensor.

The force–magnetic hysteresis curve is shown in Fig. 5(f); when different dynamic force is applied, the strain generates with speeds of 2, 50, 150, and 200 mm/min. Due to the applied force, the magnetic field initially experiences a rapid increase followed by a slower and more consistent rise. If the force is released, the magnetic field of the material gradually decreases as the force reduces from 50 to 12 N, and then rapidly drops when the force falls below 12 N. Eventually, the magnetic field returns to its initial state. As the pressing speed increases, the number of recorded data decreases. However, the data for high-speed pressing remains the same as that for low-speed pressing. Additionally, the magnetic fields at the beginning and end of the procedure remain unchanged. This means that the force–magnetic hysteresis of the semicircular hollow structure is not significantly affected by the speed of applied force. The misalignment of the two curves is due to the elastic hysteresis of the semicircular hollow structure, which can be seen in the force–strain hysteresis curve.

In summary, from the above tests resulting of the strain and magnetic field of different magnetic components, it can be seen that the semicircular hollow structure made of porous material not only enlarges the strain of the magnetic component, but also makes the magnetic component have a higher sensitivity and linearity. Besides that, this magnetic permanent not only works well as a part of the self-powered pressure sensor, but also has the potential to be used in more occasions.

B. Performance of the Sensors

The induced voltage and the applied force are displayed in Fig. 6(a). The force-detecting machine is pressing above the sensor at a speed of 350 mm/min. When the force F reaches 30 N, it remains constant for 0.04 min and then is released at a speed of 350 mm/min until the force reaches 0 N. Subsequently, the force remains constant for 0.04 min. A press and release are defined as a cycle. (F is used to indicate the maximum of the applied force, and v is used to indicate the speed at which the material deforms in the direction of the force.)

By comparing these testing results, it can be seen that the induced voltage is generated during the processes of pressure loading and releasing, and the polarity of the voltage generated during the loading and releasing processes is the opposite. The polarity of the pulses is determined by the variation of the magnetic field produced by magnetic components. The negative pulse (NP) is higher than the positive pulse (PP) for the following reason: the action of pressing and releasing is not entirely symmetrical. Another reason is the asymmetry of the semicircular hollow structure. This results in the increase and decrease of the magnetic field during pressing and releasing not following the same path, as shown in Fig. 5(f). The voltage generated by cuboid and semicircular hollow structures when $F = 30$ N and $v = 350$ mm/min is depicted in Fig. 6(b). It illustrates that the force applied to these two structures and the voltage generated by these structures in a cycle. The amplitude of the positive and NPs in two graphs is similar, but the duration of the pulses differs. When the pressure increases or decreases, the voltage appears as an NP/PP. The peak of the pulse is usually generated during the variation of pressure. The rising time of the pulse t_u is related to the speed v. This results in a smoother voltage curve for the semicircular hollow structure, and the pulses generated by the semicircular hollow structure distinctly last longer. During the release process of the force, the voltage generated by the semicircular hollow structure varies according to the manner in which pressure is applied. This phenomenon is not readily

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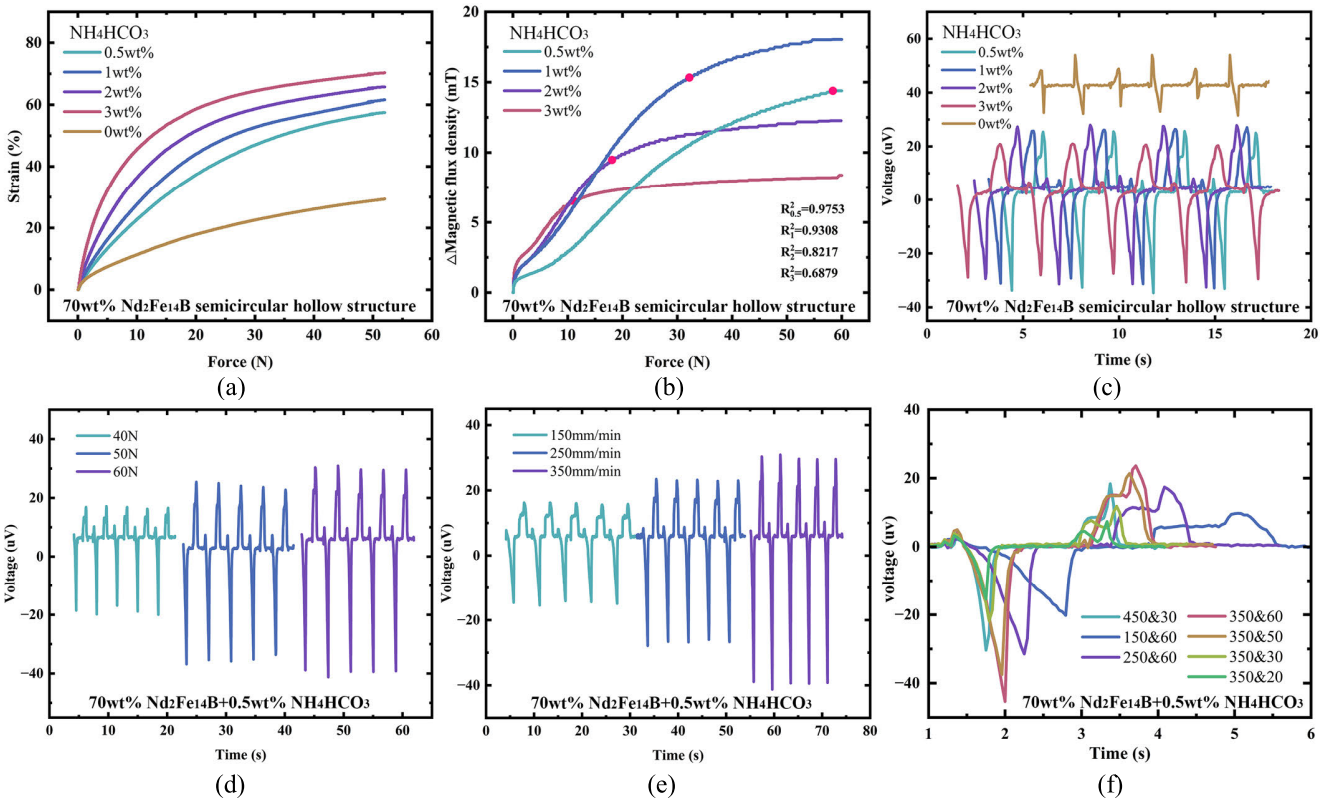

Fig. 7. Under the premise that the magnetic component is a semicircular hollow structure doped with 70 wt% $Nd_2Fe_{14}B$ and different amounts of NH_4HCO_3. (a) Relationship between the strain of the magnetic component and the applied force. (b) Changes in the magnetic fields under stress. (c) Voltage is generated by the sensor in response to applied pressure. With identical composition and doping level of magnetic components, the output voltage of the sensor is contingent upon variations in (d) applied forces and (e) speeds. (f) Output voltage waveforms of the sensor for different pressing modes.

apparent in the cuboid structure. Therefore, the following test is all based on the semicircular hollow structure. In order to display the results better, the output voltage used the display methodology is shown in Fig. 6(c).

The mechanical characteristics, magnetic field variation, and output voltage of the magnetic components doped with different quantities of NH_4HCO_3 are compared for a semicircular hollow structure containing 70 wt% $Nd_2Fe_{14}B$. According to Fig. 7(a), as the doped amounts of NH_4HCO_3 progressively rise, the material becomes more sensitive to force. The results of the test agree with the study of cuboid structure in Fig. 5(b). The component is hard without doped NH_4HCO_3. Fig. 7(b) shows the relationship between the magnetic field and the force of a material when different proportions of NH_4HCO_3 are added.

The flexibility of the magnetic component has some effect on the dynamic range of the sensor. Although the mechanical properties of the material have improved, the dynamic range of the magnetic field has decreased. The dynamic range of the magnetic field for materials doped with 2 wt% and 3 wt% NH_4HCO_3 is only 12.27 and 8.83 mT, respectively, which is much lower than the maximum dynamic range of 14.39 and 18.04 mT observed for materials doped with 0.5 wt% and 1 wt% NH_4HCO_3. This indicates that doping NH_4HCO_3 affects the magnetic properties of the magnetic components. This conclusion is the same as conducted in Table III.

Further, four magnetic components with different concentrations of NH_4HCO_3 have a high sensitivity at low force, and their sensitivity begins to decrease rapidly when the force is greater than a certain value. The reason for this is that the softer the material, the lower the force is required to get the largest strain of the semicircular hollow structure. Take materials doped with 3 wt% NH_4HCO_3 as an example. With the force range from 0 to 11.2 N, the top of the semicircular object is getting closer to the bottom before they contact. The sensor shows the highest sensitivity in this force range. In the force range of 11.2∼60 N, the top of the semicircular object touched the bottom, and the deformation of the structure becomes more difficult, which results in low sensitivity.

In summary, the deformation of the semicircular hollow structure varies under different pressure ranges. The magnetic components with 3 wt% and 2 wt% NH_4HCO_3 exhibit higher sensitivity for forces lower than 11.2 and 21.0 N. This range of forces is deemed insufficient for sensor applications because the magnetic component used as a sensor needs high sensitivity. Consequently, the semicircular hollow structure doped with 0.5 wt% NH_4HCO_3 has a stable sensitivity in the force range of 0∼60 N.

Fig. 7(c) illustrates the voltage generated by semicircular hollow structures with varying concentrations of NH_4HCO_3 under identical compression. The voltage generated by the semicircular hollow structures with 0 wt% NH_4HCO_3 is similar to that of the cuboid due to the high hardness

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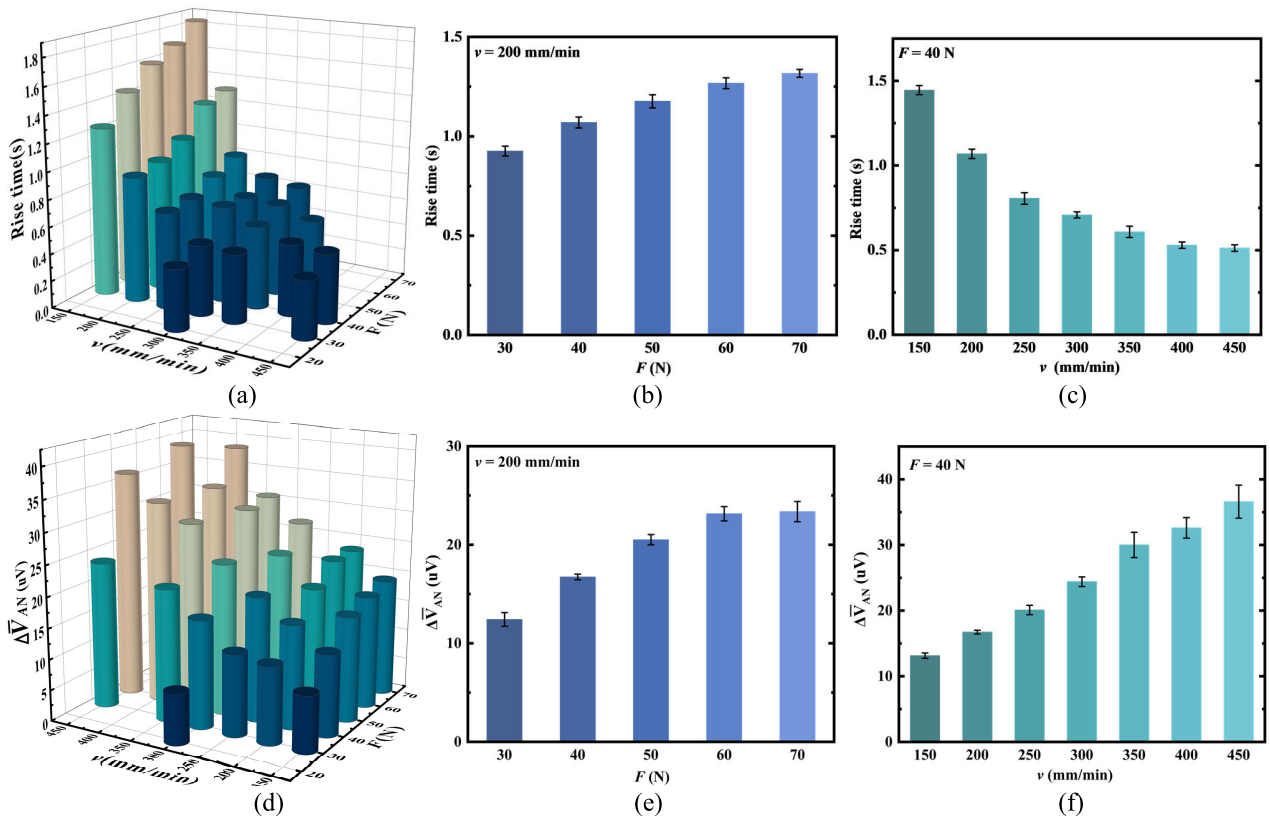

Fig. 8. (a) 3-D cylindrical diagram showing the effect of the F and v sensor applied on the rise time t_u of the output voltage by the sensor. Effect of applied F and v on the rise time t_u of the sensor output voltage. (b) Rise time t_u of pulse voltage generated in different forces F with a speed of 200 mm/min. (c) Rise time of the pulse voltage t_u generated by pressing at different speeds with a force of 40 N. (d) 3-D cylindrical diagram showing the effect of the F and v sensor applied on the negative amplitude of the output voltage $\Delta\bar{V}_{AN}$ by the sensor. (e) Negative amplitude of pulse voltage $\Delta\bar{V}_{AN}$ generated in different forces F with a speed of 200 mm/min. (f) Negative amplitude of pulse voltage $\Delta\bar{V}_{AN}$ generated in different speeds with a force of 40 N.

of the material. The amplitude of the voltage generated by a material doping 3 wt% NH_4HCO_3 is smaller than the others. In summary, softer magnetic materials are employed when greater sensitivity is needed and the harder magnetic materials are suitable for the situation that needs wider measuring ranges. In general, the mechanical properties of the material can be trimmed by varying the concentration of NH_4HCO_3. Consequently, the material doping of 0.5 wt% NH_4HCO_3 was used to fabricate the sensitive component for the following applications.

It has been deduced from (1) that the force F and speed v affect the pulse peaks. Fig. 7(d)–(f) illustrates the voltage generated by the sensor under different forces and speeds. As depicted in Fig. 7(d), under the speed of 350 mm/min, the sensor generated voltages with PP amplitudes of 10, 21.27, and 23.96 μV at F is 40, 50, and 60 N, respectively. The amplitude of the pulse signal is positively correlated with the force. When the sensor is pressed with $F = 50$ N, as shown in Fig. 7(e), the amplitude of the voltage pulse output fluctuates with speed. The sensor generated voltages with PP amplitudes of 10.14, 17.19, and 23.96 μV at speeds of 150, 250, and 350 mm/min, respectively.

As speed v increases, the duration of the voltage pulse decreases. This suggests that speed v can influence both the amplitude and waveform of the voltage pulse signal. As shown in Fig. 7(f), the initial voltage of the voltage graph is unified, and one of the waveforms is extracted under various conditions for comparison. For convenience, the phrase "speed and force" was used to express different ways of pressing. The impact of force F and speed v on the amplitude $\Delta\bar{V}_{AN}$, and the maintained time of a voltage pulse are studied. The variation ratio that the voltage rises to its peak at the same speed v is almost the same. The voltage will ascend to its peak slowly under slow pressing speed. It means that the speed v influences the rise time t_u of the voltage pulse. The time for the pulse to decrease, t_d, is relatively constant and is around 0.2 s, regardless of the speed and force applied during the pressing procedure.

Experiments were carried out to elucidate the correlation between the input force F, speed v, and the output pulse amplitude $\Delta\bar{V}_{AN}$ and rising time t_u of the sensor. The results are shown in Fig. 8. Fig. 8(a) depicts the relationship between F, v, and t_u with a 3-D cylindrical diagram. Fig. 8(b) and (c) are extracted from Fig. 8(a) to better demonstrate the relation of the rising time and force as well as force loading speed, which are more intuitive. The variable t_u exhibits a positive correlation with F while v remains constant. When the variable F remains constant, the variable t_u exhibits a negative correlation with the variable v. This is because the rise time t_u corresponds to the time taken for the force to load. According to Fig. 5(e), the force on the sensor F and deformation L (unit: mm) is in accord with the unique function and positively

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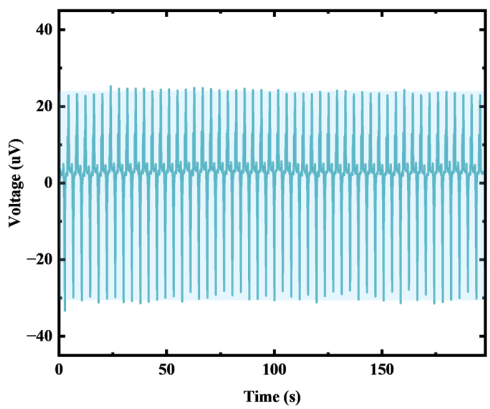

Fig. 9. Repeatability of the pressure sensor with 50 load–unload cycles applied at a force of 40 N and a speed of 350 mm/s.

correlated, written as

$$L = f(F). \tag{2}$$

Also, L and v (unit: mm/min) are in accord with the functional $t = L/v$, and t describes the time the material taken to form strain. According to the above equations, it can be concluded that

$$t = f(F)/v. \tag{3}$$

The mean absolute percentage error (MAPE) between the true value t and the measured value t_u is 0.09. The reason for this error is that it takes time for the force-detecting machine to accelerate/decelerate after it reaches the specified speed v. According to (3), the conclusion of Fig. 8(a)–(c) can be deduced.

Fig. 8(d)–(f) demonstrates that the sensor output is influenced by both the force F and the speed v. It can be seen that $\Delta\bar{V}_{\mathrm{AN}}$ is positively correlated with F when v is constant. When F is constant, $\Delta\bar{V}_{\mathrm{AN}}$ is positively related to v. This conclusion is the same as the results shown in Fig. 7.

Fig. 9 shows the output voltage of the sensor over 50 cycles under the force of 40 N and 350 mm/s. The graph shows a PP amplitude ranging from 19.78 to 22.24 μV, with a coefficient of variation (CV) of 3.03%. The NP amplitude ranges from 30.86 to 36.35 μV, with a CV of 3.38%. Little dispersion proves the good repeatability of the sensor.

C. Construction of a Sensor Test Circuit

An embedded testing system is designed to realize the portable testing of the sensor. A schematic of the test system is shown in Fig. 10(a), which includes a microcontroller, a digital multimeter, an amplifier, and a sensor. A low-power amplifier (AD8421, Analog Devices) is used in the amplifier circuit to amplify the sensor. The amplified voltage is sent to the microcontroller STM32 and then analyzed. Fig. 10(b) shows the experimental setup for the application test of the sensor with semicircular hollow. The sensor is tested by manual finger pressing rather than the mechanical loading method. As can be seen from Fig. 10(c), the digital multimeter displays the voltage read when pressing the sensor. The system can display the real-time voltage V and the number of compressions NUM in a certain amount of time. It can additionally calculate the real-time amplitude values of the PP and NP signals, $\Delta\bar{V}_{\mathrm{AP}}$ and $\Delta\bar{V}_{\mathrm{AN}}$, showing the maximum amplitude values of the PP and NP signals ${}_{\mathrm{MAX}}\Delta\bar{V}_{\mathrm{AP}}$ and ${}_{\mathrm{MAX}}\Delta\bar{V}_{\mathrm{AN}}$ since the beginning of the test. This application provides fresh perspectives into the utilization of magnetic sensors in wearable technology.

D. Sensor Application

The semicircular hollow structure sensor can generate different voltage waveforms in multifarious applications where there is only one detective point. Three distinct finger pressing methods are employed to demonstrate the output voltage waveform (unconnected amplifier) in Fig. 11. Fig. 11(a) displays the output voltage of the sensor during four consecutive presses followed by four consecutive releases. The output voltage of the sensor when pressed and released quickly and slowly is displayed in Fig. 11(b) and (c), respectively. As a result, the aforementioned three voltage waveforms can be utilized to test the variations in force, speed, and interval duration.

Compared with the semicircular hollow structure, the sensor array features a plurality of force detection points. Assembling with a conductive coil, the sensors with semicircular hollow array structures have the ability to recognize the number of points with force applied. Fig. 12 illustrates the voltage output of the sensor with a semicircular hollow structure array with nine force detection points. "Count" in the table means the number of points that force is applied. Given that there are nine sensor cells in the array, the maximum value for the "Count" variable is 9. The NP indicates that a point in the array is pressed, and the PP indicates that a point in the array is released. The voltage pulse amplitude could be affected by the pressing force F and the speed v, as well as the pressing position. If the pressing position approaches the center, the amplitude of the output voltage increases. Although it is the polarity of the output voltage used to convey number information in this experiment, the pulse amplitude of the output voltage has the potential to be used to convey location information of the pressing points by optimizing the coil pattern.

The cylindrical array is similar to the cilia structure [33]. Pressing the cylindrical structure to one side generates a variation in the magnetic field. Its detection points outnumber those of the semicircular hollow arrays, making it more suitable for large-area force detection. In wearable experiments, enameled wire is used instead of copper foil as a spiral conductive coil for testing. In Fig. 13, the cylinder array sensor is applied at the wrist. It is observable that a steady voltage waveform is generated when the wrist moves in an upward and downward motion.

When the sensor is used as wearable electronics to detect a wide range of human movements, it can be divided into two parts. As shown in Fig. 14(a), the induction coil and magnetic component were placed on the lower arm and the upper arm, respectively. There are four pulses generated during telescoping in a cycle. The first peak of the NP and the second peak of the PP are generated by the proximity of the magnetic material to the coil, and the second peak of the NP and the

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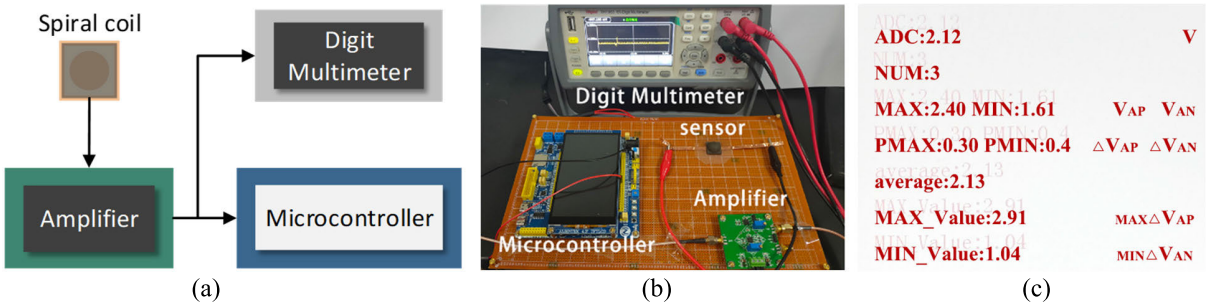


Fig. 10. (a) Schematic of the test wiring. (b) Experimental setup to achieve sensor portability testing. Digital multimeters, amplifiers, and microelectromechanics all contain the necessary equipment. The sensor is tested by manual finger pressure rather than mechanical means. (c) Microcontroller screen displays test voltage output.

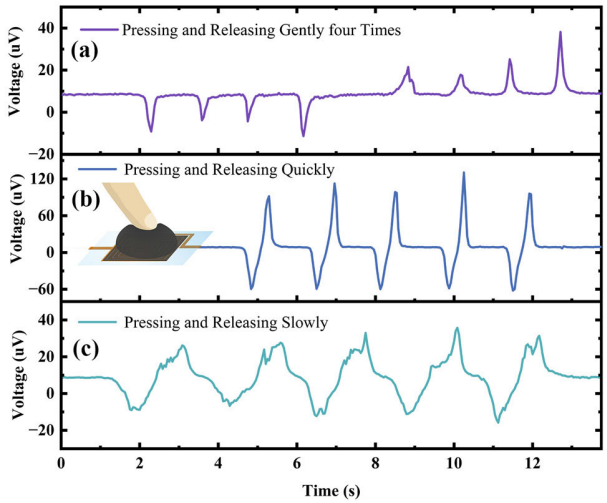
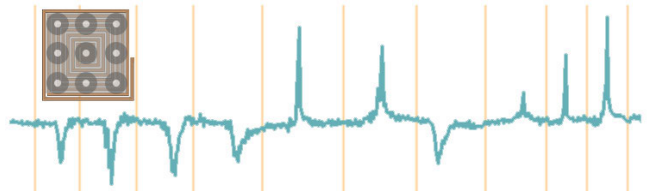


Fig. 11. Finger pressure semicircular hollow sensor application. (a) Pressing and releasing gently. (b) Pressing and releasing quickly. (c) Pressing and releasing slowly.



	--	NP	NP	NP	NP	PP	PP	NP	PP	NP	PP	PP	PP
calculate	--	+1	+1	+1	+1	-1	-1	+1	-1	+1	-1	-1	-1
Count	0	1	2	3	4	3	2	3	2	3	2	1	0

Fig. 12. Semicircular hollow array is used as a pressing point counter.

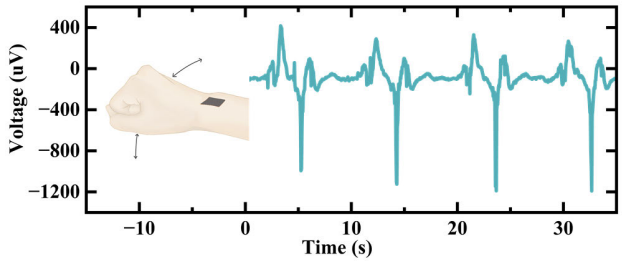


Fig. 13. Application of cylindrical array sensors in the wrist.

first peak of the PP are generated by the magnetic component being pressed. The voltage in the diagram is divided into two parts: the slow contraction/stretch of the arm produces the first part of the voltage, while the fast stretch/contraction

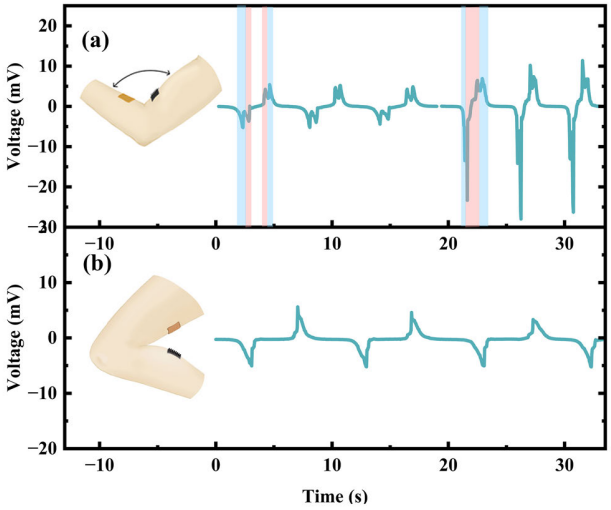


Fig. 14. (a) Sensor is demonstrated in the arm and (b) leg position for noncontact–contact application.

of the arm produces the second part. As compared to fast contraction/stretch, the voltage during slow contraction/stretch is found to have a smaller pulse amplitude. As demonstrated in Fig. 14(b), the induction coil and magnetic component are positioned inside the thigh and calf, and they are capable of producing a steady voltage while the leg is bent and extended. Due to the large force applied by the leg movement, the pulse signals here are generated mainly due to the swift proximity of the magnetic material to the coil. Compared with the arm test, only one NP/PP signal can be detected in one contraction/stretch.

Most of the pressure sensors based on flexible magnetic materials utilize the magnetoelastic effect. Table IV shows the comparison of presently reported pressure sensors based on magnetic flexible materials from the view of their working principle, structure, and function. Compared with other sensors, this article uses a purely flexible semicircular hollow material that can be adapted to scenarios with high elastic demands, and the magnetic field of the semicircular hollow magnetic structure varies nearly linearly with pressure, which means that it has the potential to be used as a sensor without a coil. Biological organisms are sensitive to the speed of force application. When the same force is applied to a human body at different speeds, the faster one may be harmful [34], [35]. In addition, the speed of force application

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TABLE IV
COMPARISON OF DIFFERENT PRESSURE SENSORS BASED ON MAGNETIC MATERIALS

Structural features		Magnetic materials	Flexible substrate	Working Principle	Functions	References
Embedded structure	Flexible magnetic material embedded in nonmagnetic flexible substrate	$Nd_2Fe_{14}B$	PDMS	Magnetoelastic effect	3D force data from head impacts detection	[11]
	Rigid magnet embedded in flexible substrate	Neodymium magnet	silicone rubber	The approach of the magnet changes the magnetic field of the target	3D force vector sensing, robotic tactile sensing	[37]
	Rigid cylindrical magnet embedded in flexible substrate	Permanent magnet	silicone rubber	Electromagnetic induction, piezoresistive effect	Real-time tri-axis forces measurement, slip detection	[14]
Ciliated structure	Double-layer magnetic cilia	$Nd_2Fe_{14}B$	PDMS	Magnetoelastic effect	Airflow perception, surface roughness characterization	[24]
	Flexible magnetic material with micro-cilia array combined with spiral electric coil	$Nd_2Fe_{14}B$, Carbonyl iron	PDMS	Electromagnetic induction, Magnetoelastic effect	Pressure detection, recognition of micro-scaled 3D morphologies, self-powered electronic skin, underwater communication, etc	[13]
Film layer structure	Flexible magnetic material with porous layer combined with liquid metal spiral coil	$Nd_2Fe_{14}B$	silicone rubber	Giant magnetoelastic effect, Electromagnetic induction	Wearable devices, powering wearable bioelectronics	[27]
Semicircular hollow structure	Flexible magnetic material with semicircular hollow structure combined with spiral electric coil	$Nd_2Fe_{14}B$	PDMS	Magnetoelastic effect, electromagnetic induction	Wearable devices, dynamic force measurement, non-contact measurement	Our work

can also reflect the controlling ability of the physical body, which is important for athletes [36]. Therefore, in this article, not only the relationship between the force F and output voltage amplitude $\Delta\bar{V}_A$ is analyzed, but also the pressing speed v and pulse rise time t_u. In general, utilizing the magnetoelastic effect and magnetic induction, a new application of magnetic sensors is developed and shows great potential to be used in a broader field.

IV. CONCLUSION

In this work, a self-powered pressure sensor was developed. Innovations were mainly realized in the structure and application of the sensor. The sensor utilizes a magnetic flexible material combined with a square spiral coil. An investigation was conducted to figure out the influence of different doping concentrations on the mechanical properties of the magnetic component. Hollow, semicircular, and porous structures are designed to improve the sensitivity of the sensor. The magnetic field variation underneath the semicircular hollow magnetic structure in the force range of 0∼50 N was found to be four times greater than that of the cuboid. Moreover, for semicircular hollow structure, the force–magnetic field curve is nearly linear. The PP and NP signals generated by the sensor can quickly determine the pressure or tension, and the amplitude of the pulse signal is determined by the speed of the force applied, which means that the rise time and amplitude of the voltage pulse can be both affected by the force F and the speed v, and the fall time of the pulse is maintained at 0.2 s. As an application, portable test systems are built to detect the maximum amplitude of the voltage signal, and the number of presses during a certain time frame. The semicircular hollow array sensor can be used as a counter to record the number of positions currently under pressing. The cylindrical array sensor is utilized for noncontact testing and can be used to record the movement of the arm or thigh. Future research will optimize the sensor structure aiming to realize self-powered pressure sensors based on flexible magnetic materials with a larger range and better linearity and also will expand the application fields of flexible magnetic sensors, such as monitoring the loading speed of force.

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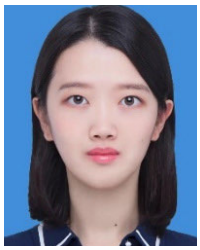

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